Proxy-Enhanced Prototype Memory Network for Weakly Supervised Hyperspectral Target Detection

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***Abstract*— Hyperspectral target detection (HTD) holds significant promise in numerous earth vision applications, yet it encounters challenges in acquiring high-quality prior target signatures, capturing target spectral variability, and dealing with sample imbalance. To address these issues, we propose a weakly supervised solution, the proxy-enhanced prototype memory network (PE-PMN), for HTD tasks. It relies solely on region-level weakly labeled data, eliminating the need for strict prior target knowledge (e.g., handcrafted target signatures or pixel-level annotations). To fully describe target variations and background diversity, two memory prototype networks are introduced to extract, store, and retrieve prototypes of targets and backgrounds, providing comprehensive spectral information. Additionally, a proxy-based enhancement approach is incorporated to enrich the prototypes in the memory banks and boost the separation between target and background features. To mitigate sample imbalance in the PE-PMN, we developed the bag mix-up (BMU) strategy based on the unconstrained linear mixture model (ULMM) to construct a sufficient training dataset. Experimental results on three simulated datasets and three real datasets demonstrate that the proposed PE-PMN significantly outperforms other competitive weakly supervised HTD methods.**

***Index Terms*— Hyperspectral, prototype memory network (PMN), target detection, weakly supervised learning (WSL).**

I. Introduction

HYPERSPECTRAL imaging is a prominent technology for capturing detailed spectral information across a wide range of wavelengths, offering abundant insights into the complex composition and nuanced characteristics of materials in the scene [1]. Hyperspectral target detection (HTD) has garnered significant attention as a prominent research direction in the hyperspectral image processing field. It leverages the spectral characteristics of individual pixels in an HSI to determine whether the pixel belongs to the target of interest or not.

HTD methods can be broadly classified into two categories: classic HTD and anomaly HTD, depending on whether the target's prior spectral signature is available. Anomaly HTD techniques focus on identifying anomalous targets that differ from the background, without relying on the target's spectral characteristics, which makes them unsuitable for detecting specific target classes. In contrast, when the target signature is known, classic HTD methods typically yield more accurate and favorable results for the class of interest.

However, the target signatures provided by spectral libraries, hand-held spectrometers, or manual selection always affect the stability of detection performance for two reasons as follows.

1) There are significant disparities between the spectra in real-captured HSIs and those in the spectral library, primarily due to variations in the acquisition equipment and the collection environment. Specifically, different sensors may have distinct band configurations and varying spectral ranges/resolutions, leading to potential spectral variations. The collection of HSIs is susceptible to various atmospheric conditions. For instance, scattering and absorption of light by atmospheric particles or moisture can alter the spectral signatures, introducing deviations from those in the spectral library.
2) HSIs typically demonstrate subpixel phenomena and targets of interest may occupy only a few pixels due to the relatively limited spatial resolution of hyperspectral sensors. The spectral responses of targets and backgrounds often exhibit mixing effects, making it hard to precisely identify and manually select target signatures from the surrounding background.

To cope with these difficulties, we focus on the rigid spectral prior-free solution for HTD tasks and aims to develop an effective detector using region-level labeled training data in a weakly supervised learning (WSL) manner, which is more flexible and easily applicable to real-world remote observations. Previous work has primarily tackled this problem under the multiple-instance learning (MIL) framework. MIL is a subfield of WSL. Instead of receiving samples with precise pixel-level labels, the training data in MIL is organized into bags with bag-level annotations, where each bag contains multiple instances. In practice, it is feasible to obtain data bags with bag-level labels. Specifically, experts can easily delineate regions that include or exclude the target (i.e., *positive* or *negative* bags, respectively) by leveraging GPS information, conducting visual observations, or utilizing high-resolution auxiliary images.

HTD techniques under the MIL framework can be broadly categorized into two primary families, that is, target characterization (TC)-based methods and multiple-instance neural network (MINN) category. TC-based methods adaptively learn target and/or background concepts, eliminating the reliance on strict spectral priors such as pure target spectra, as shown

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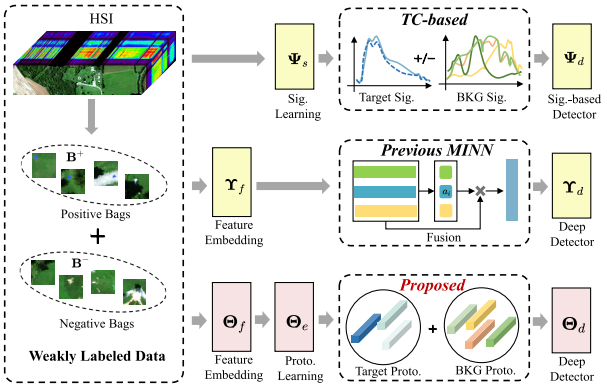

Fig. 1. Fundamental differences between the proposed PE-PMN and the previous TC-based methods [2], [3], [4], [5], [6] as well as previous MINN methods [7], [8], [9], [10]. From the feature learning perspective, the TC-based methods extract one or a set of target (and background) spectral signatures from weakly labeled data bags. Previous MINN methods for HTD commonly employ a neural network parameterized attention structure to identify potential target positions and fuse spectral features. In contrast, the proposed PE-PMN employs two learnable prototype memory networks (PMNs) with dynamic capacity and dimensionality to explicitly and adaptively capture target subtypes and background diversity.

in Fig. 1. From the perspective of the types and quantities of adaptively learned concepts, these methods can be divided into four categories: single-target (ST) concept [2], [11], [12], [13], ST concept and multiple background (STMB) concepts [3], Multiple-target (MT) concepts [4], [5], and MT concepts and multiple background concepts (MTMB) [6]. For instance, Zare et al. [2] presented multiple-instance adaptive coherence/cosine estimator (ACE)/spectral matched filter (SMF) (MI-ACE and MI-SMF) to estimate a discriminative target signature from imprecisely labeled HSI data by maximizing the ACE/SMF response of at least one pixel in each positive bag while suppressing the responses of all pixels in negative bags. Expanding on this ST-based framework, the STMB-based approach, multiple-instance subpixel ACE (MI-SPACE) [3], introduces an explicit background modeling for HTD. This method promotes the additional learning of a set of background endmembers to effectively separate the background components within subpixel targets and improve the detection response of subpixel targets. Differing from STMB-based methods, MT-based methods [4], [5] focus on learning multiple spectral signatures to capture the diversity of targets. MTMB-based methods, such as MI-HE [6], simultaneously learn spectral dictionaries for both targets and backgrounds, which improves the discrimination and detection performance in complex scenarios.

Unlike TC-based methods, MINN methods [7], [8], [9], [10] for HTD commonly employ a neural network parameterized structure to identify potential target positions within bags, in conjunction with a binary classifier. Attention-DMIL [7] is the most representative MINN method, which could find key instances within a bag/region through a flexible and adaptive MIL pooling operation. Based on this, Wang et al. [8] proposed the Max-attention MINN, which focuses primarily on the sample with the highest attention weight in each data bag. Jiao et al. [10] considered the spatially sparse distribution prior of targets and presented an L_1 sparsity-regularized attention MINN (L_1-attention MINN). Yang et al. [14] proposed a multiple-instance metric learning neural network (MIML-Net) with a metric-based regularizer, which promotes a discriminative detector by constraining target and background representations to two learnable compact clusters with distinct centroids in feature space. Despite the significant achievements of previous contributions, there are still some concerns that require appropriate addressing or further improvement.

1) HTD always faces the extreme training sample imbalance due to that targets of interest are usually rare and exhibit a low probability distribution in an observed scene. Sample imbalance poses several disadvantages for HTD, such as biased model performance and reduced generalization ability, especially for deep models. However, bag-level data imbalance for weakly supervised HTD has been largely overlooked.
2) Due to the limited spatial resolution of hyperspectral sensors and the inherent complexity of surface material distributions. Target spectral signatures mix with surrounding materials to varying degrees, leading to strong intraclass variability and the emergence of numerous subtypes. TC-based algorithms only learned one or a few signatures to describe targets, which has limited descriptive power to represent complex spectral interactions and nonlinear mixing effects.
3) Effectively suppressing heterogeneous backgrounds while precisely detecting and emphasizing targets is always a significant and formidable challenge, especially for weakly labeled HSI data.

Given these challenges, we outline a PE-PMN to solve the weakly supervised HTD problem. The PE-PMN framework integrates two key components: dynamic prototype banks (PBs), which extract, store, and retrieve diverse target and background concepts from weakly labeled data, and a proxy-based enhancement (PE) module, which enhances the discriminability of these features and boosts the target–background separation. Together, these components enable the model to more effectively capture variations within the data and improve overall detection performance. Moreover, a bag mix-up (BMU) approach is exploited to mitigate the sample imbalance phenomenon by creating sufficient synthetic training bags under the unconstrained linear mixture model (ULMM) framework. In summary, the innovations of this work are organized as follows.

1) To provide sufficient training data bags for deep models and handle the data imbalance problem, a bag-level data augmentation procedure BMU inspired by the ULMM for weakly supervised HTD is first proposed in this article.
2) To fully characterize the subtypes of targets arising from spectral mixing and even backgrounds, we construct two PBs to store and retrieve target and background concepts, as illustrated in Fig. 1. The PB incorporates an explicit memory component with adjustable capacity, where each prototype represents distinct subtypes of mixed targets or backgrounds, ensuring that variations

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caused by spectral mixing are effectively captured. Even highly mixed spectra can be accurately assigned to the correct category through comparison with the most similar stored prototypes.

3) For emphasizing targets and suppressing backgrounds, we introduced a PE strategy. By introducing proxies for memory-augmented features, PE improves the discriminability of target and background features, leading to more compact representation learning. Besides, the proxies are global in nature, which can capture information beyond the scope of individual batches.
4) Leveraging PBs and PE, a weakly supervised solution PE-PMN is proposed for weakly supervised HTD. To the best of our knowledge, the PE-PMN is the first work that enables the incorporation of PBs to describe hyperspectral subtypes for weakly supervised HTD. It relaxes the requirement for prior signatures and allows region-like training samples.

The rest of this article is arranged as follows. Section II provides a comprehensive review of HTD and discusses various sample imbalance solutions for HTD. In Section III, we describe the proposed BMU strategy and PE-PMN architecture, respectively. Then, we prove the superiority and effectiveness of the proposed methods through extensive empirical experimentation and rigorous data analysis in Section IV. Finally, Section V provides a concise summary of the key contributions of this study and outlines promising avenues for future research.

II. Related Work

A. HTD: From Shallow to Deep

In the early stages of HTD development, it is treated as a hypothesis-testing problem, which is solved by generalized likelihood ratio testing (GLRT), such as SMF [15] and ACE [16]. Subsequently, the signal decomposition-based methods constantly sprang up [17]. These approaches consider that the spectrum can be physically decomposed as several components with different fractional abundances, that is, orthogonal subspace projection (OSP) [18], and low-rank and sparse matrix decomposition (LRaSMD) [19]. The linear filtering-based HTD algorithms also gain great success. The most representative one of this family, constrained energy minimization (CEM) [20], demonstrates remarkable detection performance by effectively enhancing the targets and suppressing the backgrounds through a finite impulse response (FIR) filter. Chang and Ren [21] extend CEM into a multitarget version, termed linear constrained minimum variance (LCMV). Other extensions, hierarchical CEM (hCEM) [22] and Bayesian CEM [23], also show promising detection performance.

With decades of development, many machine learning (ML)-based algorithms are developed for HTD tasks, especially kernel-based ML, and sparse and collaborative representation [24]. Dong et al. [25] proposed an HTD method based on asymmetric weighted logistic metric learning (AWLML), which demonstrates great detection accuracy in scenarios with imbalanced target–background distributions. Cheng and Wang [26] proposed a decomposition model (DM) with background dictionary learning (BDL), effectively separating background, target, and noise components and demonstrating robustness against inaccurate prior target information. Similarly, to address the issue of insufficient target pixels in building sparse representation models, Zhu et al. [27] presented a single-spectrum-driven binary-class sparse representation for HTD (SSBSTD) with a global overcomplete background dictionary. As for the time-series problem in HTD, Zhao et al. [28] offered a solution that incorporates spectral perception and spatial–temporal tensor decomposition, thereby leveraging the correlation among backgrounds and the temporal continuity of the targets. Moreover, several methods address the problem from a Bayesian perspective. For instance, Matteoli and Theiler [29] presented a comprehensive Bayesian framework for HTD, showing competitive performance compared to GLRT-based methods.

In recent explorations, many data-driven deep learning (DL)-based approaches have been developed. Li et al. [30] presented a distribution-free two-step subspace projection (TSP) detector that reduces interference via a hierarchical denoising autoencoder (HDAE). To cope with the sample imbalance problem in deep models, two-stream convolutional network-based target detector (TSCNTD) [31] is proposed to extract abundant spectral features through the two-stream neural networks, coupled with a spectral synthesis method to construct a wealth of training samples. Some scholars try to encourage the separation between targets and backgrounds and reduce the intraclass variations with the help of deep metric learning. For instance, Siamese [32], [33] and triplet [34] architectures are devised into the detector training process to regulate a discriminative feature space. Besides, Gao et al. [35] proposed a novel approach for HTD, termed spectral aggregation and separation network (SASN), which distinguishes the spectral similarities and differences simultaneously to effectively discern subtle variations in spectral patterns between targets and backgrounds. Furthermore, some generative models, such as the generative adversarial network (GAN) [36] and the variational autoencoder (VAE) [37], also achieved great success in extracting spectral features, contributing to the development of HTD.

When one or a set of high-quality target spectral signatures are provided, almost all of the aforementioned methods can achieve superior detection performance. However, existing techniques struggle to ensure reliable target spectrum selection. Therefore, weakly supervised target detection algorithms without a specific spectral prior are appealing.

B. Sample Imbalance Solutions for HTD

Sample imbalance is a common challenge in HTD, where the number of target pixels is often much smaller than that of background pixels. This disparity can lead to biased models that favor background regions, resulting in poor detection performance for rare targets. Over the years, various strategies have been proposed to address this imbalance, primarily focusing on sample re-weighting (SR) and data augmentation (DA).

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The SR solutions tend to assign higher weights to the underrepresented class (target pixels) to prevent the model from ignoring them during training. These methods dynamically adjust the importance of different classes according to class frequency [10], [30], or detection difficulty [33]. While SR methods have proven effective in both traditional and weakly supervised HTD tasks, DA is often preferred for its ability to allow seamless integration into existing detection models without requiring intricate loss function tuning. However, many classic DA techniques, such as rotation, scaling, and mirroring, focus on spatial transformations and neglect the spectral dimension, which is crucial for hyperspectral images. To address this issue, the target band random mask (TBRM) [35] generates augmented target samples by applying random zero-masking to specific spectral bands of a target signature. The target band random substitution (TBRS) [38] replaces the randomly selected band of a target signature with the average value of all bands to be substituted. In [31], target samples are synthesized through a linear combination of the target prior and a subset of typical background samples. The existing spectral DA methods focus primarily on pixel-level data generation, making them less suitable for weakly supervised HTD methods (e.g., MINN techniques [8], [10]) that require bag-level inputs. While pixel-level generated samples can be aggregated into bag-level samples, these DA methods still depend on target spectral priors, which are not available in MIL settings. Thus, we propose a more flexible bag-level DA method BMU, which can generate sufficient hyperspectral training bags using linear interpolation between two randomly selected and shuffled positive and negative bags without a rigid target prior and predefined background endmembers.

III. Proposed Methods

A. Problem Formulation

The training HSI data is formulated as $\mathbf{X} = [\mathbf{x}_1, \ldots, \mathbf{x}_{N_{\text{HSI}}}] \in \mathbb{R}^{d \times N_{\text{HSI}}}$, where d denotes the number of spectral band and N_{HSI} represents the total number of HSI pixels. We group the HSI data into multiple bags $\mathfrak{B} = \{\mathbf{B}_1, \ldots, \mathbf{B}_{N_{\text{B}}}\}$ with bag-level labels $\mathfrak{L} = [L_1, \ldots, L_{N_{\text{B}}}]$, where N_{B} is the total number of data bags. $L_i \in \{0, 1\}$ indicates whether the bag $\mathbf{B}_i$ contains target pixels. The data bags $\mathfrak{B}$ can be separated into two subsets, that is, positive bag subset $\mathfrak{B}^+$ with $L = 1$ and negative bag subset $\mathfrak{B}^-$ with $L = 0$. In practice, $|\mathfrak{B}^+| \ll |\mathfrak{B}^-|$. For each bag $\mathbf{B}_i$ in $\mathfrak{B}^+$, there is at least one target instance in the bag, but the specific location l_{ij} is unknown, that is, $L_i = 1$ if $\exists l_{ij} = 1$. For a negatively labeled bag, there are no target pixels, that is, $L_i = 0$ if $\forall l_{ij} = 0$. The goal is to build an effective hyperspectral target detector using weakly labeled data bags. The well-trained detector should be capable of handling HTD tasks that involve both bag-level and instance-level inputs.

B. Bag Mix-Up

In practice, targets of interest are usually rare in the scene and exhibit a low-probability distribution, while background pixels are commonly prevalent. Thus, acquiring an equivalent amount of target and background data bags proves to be challenging. This sample imbalance phenomenon can pose challenges for detection model building, such as 1) the detector tends to predict the majority class (background), leading to biased performance toward targets and 2) the DL-based detector training on an imbalanced dataset may exhibit poor generalization in identifying the targets with low occurrence probabilities. Motivated by these issues, an effective bag augmentation routine for generating sufficient training datasets is essential.

To address this, we employ the ULMM to represent hyperspectral pixels. The ULMM assumes that each spectral vector can be expressed as a linear combination of multiple endmembers with varying fractional abundances, such as $\mathbf{x}_{mj} = \mathbf{t}\boldsymbol{\alpha}^{\mathbf{t}}_{mj} + \mathbf{E}\boldsymbol{\alpha}^{\mathbf{E}}_{mj} + \mathbf{I}$ and $\mathbf{x}_{nj} = \mathbf{t}\boldsymbol{\alpha}^{\mathbf{t}}_{nj} + \mathbf{E}\boldsymbol{\alpha}^{\mathbf{E}}_{nj} + \mathbf{I}$, where $[\mathbf{t}\ \mathbf{E}]$ denotes the collection of target and background endmembers. $\boldsymbol{\alpha} = [\boldsymbol{\alpha}^{\mathbf{t}}; \boldsymbol{\alpha}^{\mathbf{E}}]$ represents the corresponding abundances. $\mathbf{I}$ is the inevitable additive noise signal (e.g., Gaussian noise).

Based on this, we propose the BMU, which constructs the virtual positive training bags using linear interpolations as follows:

$$\hat{\mathbf{x}}_{ij} = \delta \mathbf{x}_{mj} + (1 - \delta)\mathbf{x}_{nj} \tag{1}$$

$$\hat{\mathbf{B}}_i = \left[\hat{\mathbf{x}}_{i1}, \ldots, \hat{\mathbf{x}}_{iN_i}\right] \tag{2}$$

where $\mathbf{x}_{mj}$ and $\mathbf{x}_{nj}$ are the jth instances in $\tilde{\mathbf{B}}_m$ and $\tilde{\mathbf{B}}_n$, respectively. $\tilde{\mathbf{B}}_m = \text{Shuffle}(\mathbf{B}_m)$ and $\tilde{\mathbf{B}}_n = \text{Shuffle}(\mathbf{B}_n)$ are two randomly selected data bags after bag shuffle operation $\text{Shuffle}(\cdot)$. The bag shuffle operation is performed to increase the diversity of the mixtures $\hat{\mathbf{x}}_{ij}$. Moreover, δ is a random factor sampled from the Beta distribution $\mathcal{B}(\beta, \beta)$. It can be found that the mix-up virtual example $\hat{\mathbf{x}}_{ij}$ still follows the ULMM model $\hat{\mathbf{x}}_{ij} = \mathbf{t}\boldsymbol{\alpha}^{\mathbf{t}}_{ij} + \mathbf{E}\boldsymbol{\alpha}^{\mathbf{E}}_{ij} + \mathbf{I}$, where $\boldsymbol{\alpha}^{\mathbf{t}}_{ij} = \delta(\boldsymbol{\alpha}^{\mathbf{t}}_{mj} + \boldsymbol{\alpha}^{\mathbf{t}}_{nj})$ and $\alpha^{\mathbf{E}}_{ij} = (1 - \delta)(\boldsymbol{\alpha}^{\mathbf{E}}_{mj} + \boldsymbol{\alpha}^{\mathbf{E}}_{nj})$.

In our investigation, while a sufficient quantity of negative bags (background) is available, positive samples (data bags containing target components) are scarce. Thus, we randomly sample one bag from both the positive subset $\mathfrak{B}^+$ and the entire hyperspectral bag collection $\mathfrak{B}$ and then utilize the BMU strategy to generate a positive bag. By repeatedly performing this operation, we can obtain a sufficient number of positive samples. This operation ensures that the generated bags contain subpixels of the target and can effectively simulate diverse background mixtures. The workflow of BMU is summarized in Algorithm 1.

C. Proxy-Enhanced Prototype Memory Network

1) Overview of the PE-PMN: We present a PE-PMN, a weakly supervised HTD technique with dual prototype memory banks (i.e., target and background) and proxy enhancement. The diagram in Fig. 2 provides an overview of the PE-PMN. The proposed PE-PMN can be divided into three procedures, including feature extraction, proxy-enhanced prototype learning, and detector building.

2) Feature Extraction: Let $\mathcal{U} \in \mathbb{R}^D$ and $\mathcal{V} \in \mathbb{R}^{2D}$ be two intermediate representation space. A target detection function $\phi(*; \boldsymbol{\Theta}) : \mathcal{X} \mapsto \mathbb{R}$ can be represented as a combination of a spectral feature extractor $\varphi(*; \boldsymbol{\Theta}_f) : \mathcal{X} \mapsto \mathcal{U}$, a feature enhancement function $\psi(*; \boldsymbol{\Theta}_e) : \mathcal{U} \mapsto \mathcal{V}$, and a target scoring

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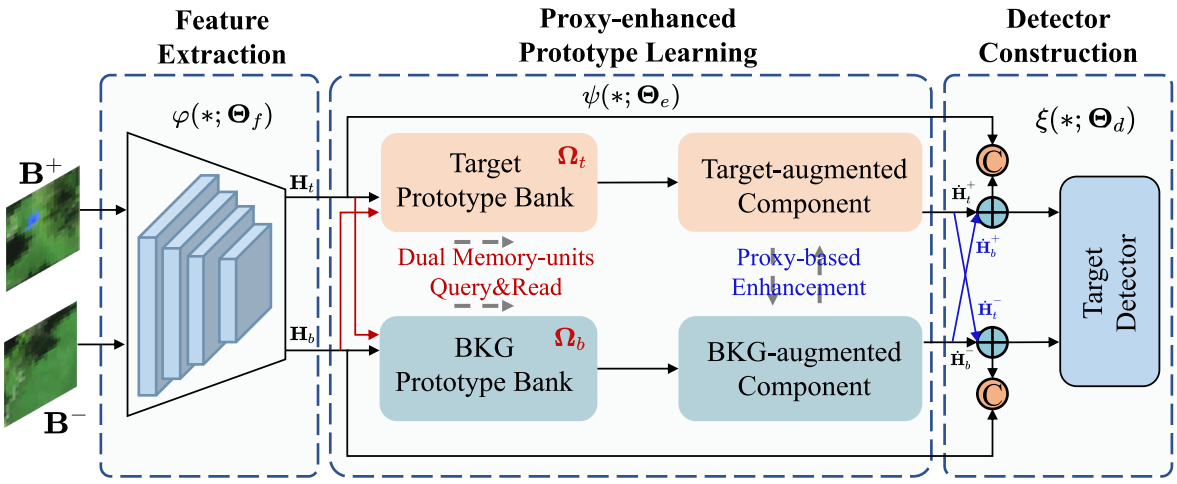

Fig. 2. Overall workflow of the proposed PE-PMN. Spectral data bags are first fed into a feature extractor for encoding. These encoded features are then separately queried against the target prototype bank and the background prototype bank, facilitating the decomposition of spectral target and background components. Next, the decomposed features undergo enhancement using a proxy-based strategy. Finally, a neural network-based detector is employed to recognize targets and suppress background.

Algorithm 1 BMU

Input: Hyperspectral bags $\mathfrak{B}$ and the positive subset $\mathfrak{B}^+$, number of generated bags N and parameter β for Beta distribution.

Output: Generated positive bag set $\mathfrak{B}_g^+$.

```
for i = 1, ..., N do
    Randomly sample two bags B_m ∈ B^+ and B_n ∈ B, respectively
    Shuffle the spectra in the selected bags B_m, B_n to obtain B̃_m, B̃_n
    Sample a random number δ from Beta distribution B(α, α)
    for j-th spectrum x_mj and x_nj in B̃_m and B̃_n do
        x̂_ij ← δx_mj + (1 − δ)x_nj
        Add x̂_ij to generated bag B̂_i
    end for
    Add generated bag B̂_i to bag set B_g^+
end for
```

function $\xi(*;\Theta_d):\mathcal{V}\mapsto\mathbb{R}$, in which $\Theta=\{\Theta_f,\Theta_e,\Theta_d\}$. Specifically, the spectral feature learner $\varphi(*;\Theta_f)$ could be designed with three convolutional modules (CMs) followed by a fully connected (FC) layer. For each CM, it comprises a 1D-CNN layer with a 1×3 kernel size, followed by a parametric rectified linear unit (PReLU), and then a max-pooling operation with a 1×2 kernel size in sequence. An FC layer is utilized to map the convolutional feature to a fixed dimension D. Empirically, M is set to 500 in our implementation. The spectral representation $\mathbf{h}\in\mathbb{R}^D$ can be formally represented as

$$\begin{aligned}\mathbf{h}&=\varphi(\mathbf{x};\Theta_f)\\&=\mathrm{FC}([\mathrm{MaxPool}(\mathrm{PReLU}(\mathrm{Conv}(\mathbf{x})))]_{\times 3}).\end{aligned}\tag{3}$$

In practice, the proposed PE-PMN method is not limited to 1D-CNN or pixel-level input. Our approach is inherently flexible and can be adapted to region-level input, allowing integration with various non-1D-CNN architectures commonly used in HSI analysis. For instance, the feature representation of a pixel $\mathbf{x}$ can be extracted from its local neighborhood $\mathcal{N}(\mathbf{x})$. In such cases, the proposed PE-PMN can be seamlessly integrated with various existing feature extraction structures, such as 2D-CNN [39], 3D-CNN [40], or hybrid models [41], followed by a feature flattening operation to obtain the region-level features, which can be expressed as $\mathbf{h}=\mathrm{Flatten}(\varphi(\mathcal{N}(\mathbf{x});\Theta_f))$. Pixel-level input can be considered a special case of region-level input when the neighborhood of a pixel is reduced to the pixel itself, that is, $\mathcal{N}(\mathbf{x})=\mathbf{x}$. In this work, we consistently used pixel-level input to ensure fair comparisons across different methods.

3) Prototype Bank Structure: In hyperspectral imaging, the spectral characteristics of targets often exhibit variability due to the influence of the acquisition environment and the limitations of the acquisition equipment. Previous hyperspectral target detectors have predominantly relied on metric learning, contrastive learning, or generative models to mitigate the spectral differences of similar targets in the feature space. However, there has been limited exploration of explicit modeling of different subtypes of targets. It restricts the ability of detectors to fully capture the inherent spectral diversity and complexity of the targets and even the backgrounds. Fortunately, recent works on memory networks [42], [43] have demonstrated the capability to characterize target diversity, offering a promising direction for more robust and comprehensive HTD.

Motivated by these contributions, two PBs, that is, $\mathbf{\Omega}_t\in\mathbb{R}^{M_t\times D}$ and $\mathbf{\Omega}_b\in\mathbb{R}^{M_b\times D}$, were established to store the target and background spectral concepts, where M_t and M_b are the number of prototypes in $\mathbf{\Omega}_t$ and $\mathbf{\Omega}_b$, respectively. By maintaining a rich and varied set of prototypes, the PBs can effectively represent the complex and overlapping spectral patterns associated with both target and background, enabling more robust and accurate spectral analysis in the presence of spectral mixing. Fig. 3 depicts the structure of the target PB, while the background PB follows a similar structure. We define a ***query*** operation, which interacts with memory banks $\mathbf{\Omega}_t/\mathbf{\Omega}_b$

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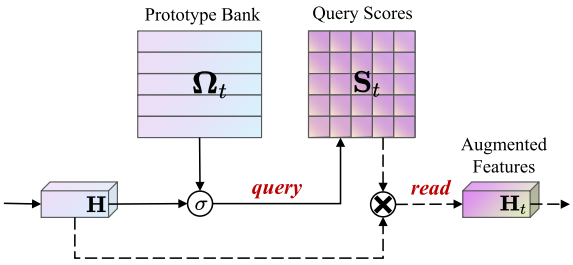

Fig. 3. Schematic of the PMN of targets, illustrating the query and read operations for extracting and retrieving target prototypes to generate the augmented features.

and produces the similarity scores between the input spectral features in a bag with the stored prototypes, as follows:

$$\mathbf{S}_t = \sigma\left(\frac{\mathbf{H}\boldsymbol{\Omega}_t^{\mathrm{T}}}{\sqrt{D}}\right), \quad \mathbf{S}_b = \sigma\left(\frac{\mathbf{H}\boldsymbol{\Omega}_b^{\mathrm{T}}}{\sqrt{D}}\right) \tag{4}$$

where $\mathbf{H} = [\mathbf{h}_1, \ldots, \mathbf{h}_N] \in \mathbb{R}^{N\times D}$ and $\sigma(*) = 1/(1+e^{-*})$ is a Sigmoid activation function to constrain query values $\mathbf{S}_t \in \mathbb{R}^{N\times M_t}$ and $\mathbf{S}_b \in \mathbb{R}^{N\times M_b}$ to $(0, 1)$. Then, a ***read*** operation is used to produce the augmented representation

$$\mathbf{H}_t = \mathbf{S}_t\mathbf{H}, \quad \mathbf{H}_b = \mathbf{S}_b\mathbf{H} \tag{5}$$

where $\mathbf{H}_t/\mathbf{H}_b \in \mathbb{R}^{N\times D}$. The prototypes in the memory block are initialized using a uniform distribution. The standard deviation (std) for this distribution is computed as the inverse square root of the prototype dimensionality $(1/\sqrt{D})$, where D is the dimensionality of each prototype. The prototypes are then initialized with random values drawn from the range $[-\text{std}, \text{std}]$. This ensures a balanced initialization, helping to avoid vanishing or exploding gradients during training. top$K$ strategy along the second dimension is applied to original query scores to determine which prototypes stored in PB are the most similar candidates to the input

$$\dot{\mathbf{S}}_t = \text{top}K_{\text{dim2}}(\mathbf{S}_t, K), \quad \dot{\mathbf{S}}_b = \text{top}K_{\text{dim2}}(\mathbf{S}_b, K) \tag{6}$$

where $K \ll \min(M_t, M_b)$. This facilitates finding approximate prototypes for highly mixed potential targets, thereby improving detection robustness against spectral mixing effects.

If we feed the data bags with different labels into the target PB and background PB, the query scores will display distinct characteristics under MIL configuration. For input feature $\mathbf{H}^-$ from a negative bag, the query scores $\dot{\mathbf{S}}_t^-$ for $\boldsymbol{\Omega}_t$ should be entirely close to $\mathbf{0}$, which implies that there are no target prototypes closed to background instances. Conversely, the query scores $\dot{\mathbf{S}}_b^-$ for $\boldsymbol{\Omega}_b$ should be entirely close to $\mathbf{1}$, which means that $\boldsymbol{\Omega}_b$ forms a relatively complete set of background pixels, allowing us to easily find multiple prototypes close to any input background instance. As for input feature $\mathbf{H}^+$ from a positive bag, the query scores $\dot{\mathbf{S}}_t^+$ for $\boldsymbol{\Omega}_t$ is not all 1, and the query scores $\dot{\mathbf{S}}_b^+$ for Ω_b is not all 0. Thus, top$K$ strategy along the first dimension is applied to determine which instances have the highest probability of being targets or backgrounds, that is, $\ddot{\mathbf{S}}_t^+ = \text{top}K_{\text{dim1}}(\dot{\mathbf{S}}_t^+, K)$ and $\ddot{\mathbf{S}}_b^+ = \text{top}K_{\text{dim1}}(\dot{\mathbf{S}}_b^+, K)$. We can utilize binary cross-entropy (BCE) loss to incorporate these constraints into the model as follows:

$$\begin{aligned}\mathcal{L}_{\text{PB}} = \text{BCE}(\dot{\mathbf{S}}_t^-, \mathbf{0}) + \text{BCE}(\dot{\mathbf{S}}_b^-, \mathbf{1}) \\ + \text{BCE}\left(\ddot{\mathbf{S}}_t^+, \mathbf{1}\right) + \text{BCE}\left(\ddot{\mathbf{S}}_b^+, \mathbf{0}\right).\end{aligned} \tag{7}$$

4) Proxy-Based Enhancement: Based on PB and query scores, we can ***read*** the augmented features $\dot{\mathbf{H}}_t^+ = \dot{\mathbf{S}}_t^+\mathbf{H}^+$ and $\dot{\mathbf{H}}_b^+ = \dot{\mathbf{S}}_b^+\mathbf{H}^+$ for positive bags and $\dot{\mathbf{H}}_t^- = \dot{\mathbf{S}}_t^-\mathbf{H}^-$ and $\dot{\mathbf{H}}_b^- = \dot{\mathbf{S}}_b^-\mathbf{H}^-$ for negative bags. Since the query values $\dot{\mathbf{S}}_t^-$ are optimized to converge to $\mathbf{0}$, the augmented feature of the negative bag on the target PB also tends to zero, that is, $\dot{\mathbf{H}}_t^- \to \mathbf{0}$. Therefore, it can be disregarded. Based on our observations, $\dot{\mathbf{H}}_t^+$ and $\dot{\mathbf{H}}_b^+$ can be considered as extracted target and background components from positive bags, while $\dot{\mathbf{H}}_b^-$ represents an augmented expression of background features from negative bags. With this insight in mind, our natural inclination is to manipulate the distances between these features to enhance the representation. Our goal is to shorten the distance between $\dot{\mathbf{H}}_b^+$ and $\dot{\mathbf{H}}_b^-$, while increasing the distance between $\dot{\mathbf{H}}_t^+$ and $\dot{\mathbf{H}}_b^+/\dot{\mathbf{H}}_b^-$. It allows for a more compact background feature space, while simultaneously creating a greater dispersion between target and background feature distribution.

Inspired by the idea of Proxy-NCA [44] and Proxy-Anchor [45], we assign two proxies to targets and backgrounds and associate them with entire data in a batch to build a PE method for augmented features, as follows:

$$\begin{aligned}\mathcal{L}_{\text{Proxy}} = \frac{1}{2}\sum_{\boldsymbol{\rho}\in\mathcal{P}}\log\left(1 + \sum_{\dot{\mathbf{h}}\in\dot{\mathbf{H}}_{\boldsymbol{\rho}}} e^{-\kappa(\cos(\dot{\mathbf{h}},\boldsymbol{\rho})-\epsilon)}\right) \\ + \frac{1}{2}\sum_{\boldsymbol{\rho}\in\mathcal{P}}\log\left(1 + \sum_{\dot{\mathbf{h}}\in\dot{\mathbf{H}}_{\boldsymbol{\rho}}^{\perp}} e^{\kappa(\cos(\dot{\mathbf{h}},\boldsymbol{\rho})+\epsilon)}\right)\end{aligned} \tag{8}$$

where $\mathcal{P} = \{\boldsymbol{\rho}_t, \boldsymbol{\rho}_b\}$ is the collection of two proxies of the target $(\dot{\mathbf{H}}_t^+)$ and the background $(\dot{\mathbf{H}}_b^+/\dot{\mathbf{H}}_b^-)$, respectively. κ is a scaling factor and $\epsilon > 0$ is a margin. $\cos(*, *)$ is the cosine similarity between two vectors. $\dot{\mathbf{H}}_{\boldsymbol{\rho}}$ represents the features associated with the proxy $\boldsymbol{\rho}$, while $\dot{\mathbf{H}}_{\boldsymbol{\rho}}^{\perp}$ denotes the features that do not belong to proxy $\boldsymbol{\rho}$. Equation (8) can be further rewritten as follows:

$$\begin{aligned}\mathcal{L}_{\text{Proxy}} = \frac{1}{2}\sum_{\boldsymbol{\rho}\in\mathcal{P}}\left[\text{Softplus}\left(\underset{\dot{\mathbf{h}}\in\dot{\mathbf{H}}_{\boldsymbol{\rho}}}{\text{LSE}}\left(-\kappa(\cos(\dot{\mathbf{h}}, \boldsymbol{\rho}) - \epsilon)\right)\right)\right] \\ + \frac{1}{2}\sum_{\boldsymbol{\rho}\in\mathcal{P}}\left[\text{Softplus}\left(\underset{\dot{\mathbf{h}}\in\dot{\mathbf{H}}_{\boldsymbol{\rho}}^{\perp}}{\text{LSE}}\left(\kappa(\cos(\dot{\mathbf{h}}, \boldsymbol{\rho}) + \epsilon)\right)\right)\right]\end{aligned} \tag{9}$$

where $\text{Softplus}(*) = \log(1 + e^*)$ denotes a smooth approximation of rectified linear unit activation, that is, $\text{ReLU}(*) = \max(0, *)$. LSE refers to the Log–Sum–Exp function, which is a smooth approximation of the max function. By minimizing $\mathcal{L}_{\text{Proxy}}$, the objective is to bring the feature distance between the proxy and its most dissimilar examples from the same class closer, while simultaneously pushing the feature distance between the proxy and its most similar examples from different classes farther apart.

5) Detector Building: During training process, we first ***read*** the augmented features $\dot{\mathbf{H}}_t$ and $\dot{\mathbf{H}}_b$ from target PB and background PB and then apply an element-wise addition to fuse those two features. The original features $\mathbf{H}$ and the fusion features are concatenated and fed into an FC layer to make the final prediction, which can be formulated as

$$\mathbf{y} = \xi(\text{FC}(\text{Concat}(\mathbf{H}; \dot{\mathbf{H}}_t + \dot{\mathbf{H}}_b)); \boldsymbol{\Theta}_d). \tag{10}$$

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Algorithm 2 Training Procedure for the PE-PMN

Input: Training bags $\mathfrak{B}$ (generated by Algorithm 1) and bag-level labels $\mathfrak{L}$. Batch size b. Learning rate lr.

Output: Parameters $\mathbf{\Theta} = \{\mathbf{\Theta}_f, \mathbf{\Theta}_e, \mathbf{\Theta}_d\}$.

Initialize parameters $\mathbf{\Theta}$. Set current iteration number $k = 0$.
repeat
  Extract spectral features $\mathbf{H}$ according to Eq. (3).
  Perform the ***query*** operation according to Eq. (4).
  Compute the PB loss $\mathcal{L}_{\text{PB}}$ according to Eq. (7).
  Perform the ***read*** operation according to Eq. (5) and Eq. (6).
  Compute the proxy loss $\mathcal{L}_{\text{Proxy}}$ using Eq. (9).
  Calculate the detection confidence according to Eq. (10).
  Compute the MIL loss $\mathcal{L}_{\text{MIL}}$ according to Eq. (11).
  Compute the total loss $\mathcal{L}$ according to Eq. (12), and perform backpropagation for parameter updating.
  $k \leftarrow k + 1$.
until Maximum training epoch reached

Algorithm 3 Testing Procedure for the PE-PMN

Input: Bag-level input $\mathbf{B}$ or instance-level input $\mathbf{x}$, the well-trained model from Algorithm 2 with parameters $\mathbf{\Theta} = \{\mathbf{\Theta}_f, \mathbf{\Theta}_e, \mathbf{\Theta}_d\}$.

Output: Bag-level prediction $\mathbf{y}$ for bag-level input or instance-level prediction y for instance-level input $\mathbf{x}$.

Extract spectral features $\mathbf{H}$ for bag-level input or $\mathbf{h}$ for instance-level input according to Eq. (3).
Perform the ***query*** operation according to Eq. (4).
Perform the ***read*** operation according to Eq. (5) and Eq. (6).
Calculate the detection confidence $\mathbf{y}$ according to Eq. (10) or y according to Eq. (13).

We select the top$K$ examples with the highest probabilities within each bag, that is, $\dot{\mathbf{y}} = \text{top}K(\mathbf{y})$, and then propagate the bag-level labels to these selected examples to build a MIL-based loss function as follows:

$$\mathcal{L}_{\text{MIL}} = \text{BCE}(\dot{\mathbf{y}}, \overrightarrow{\mathbf{y}}) \tag{11}$$

where $\overrightarrow{\mathbf{y}}$ is the propagated labels. Then, the overall loss function is defined as

$$\mathcal{L} = \mathcal{L}_{\text{MIL}} + \lambda_1 \mathcal{L}_{\text{PB}} + \lambda_2 \mathcal{L}_{\text{Proxy}}. \tag{12}$$

During the testing process, our model is flexible for instance-level input and the pixel-wise detection confidence is obtained by

$$y = \xi(\text{FC}(\text{Concat}(\mathbf{h}; \dot{\mathbf{h}}_t + \dot{\mathbf{h}}_b)); \mathbf{\Theta}_d). \tag{13}$$

By summarizing the aforementioned modules, the training procedure of the PE-PMN can be described in Algorithm 2 and the testing workflow is organized in Algorithm 3.

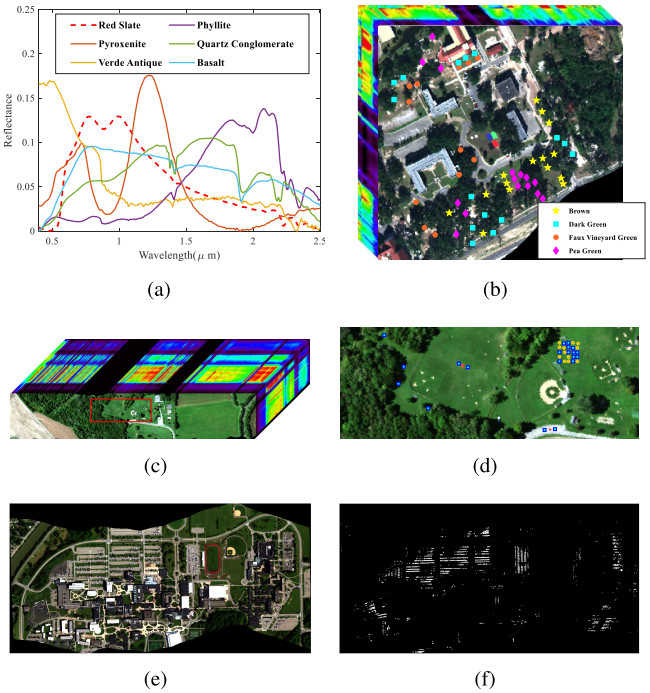

Fig. 4. (a) Six endmembers were selected from the ASTER spectral library [46]. Red Slate was designated as the target of interest, while the remaining five spectra served as background endmembers. (b) Scene of the MUULF Gulfport hyperspectral dataset [47] and the ground-truth locations of the target objects. (c) Scene of the Avon hyperspectral dataset [48]. (d) Corresponding ground-truth locations of the targets. (e) Scene captured by the AeroRIT dataset [49] over the Rochester Institute of Technology campus. (f) Locations of the labeled cars in the AeroRIT dataset.

IV. Experiments

A. Experimental Setup

1) Data Description and Task Design: The proposed PE-PMN is evaluated on three simulated hyperspectral datasets and three real-world hyperspectral datasets.

a) Simulated hyperspectral datasets: The simulated datasets are generated from the ASTER spectral library [46], as illustrated in Fig. 4(a). The target of interest for our detection task was the Red Slate spectrum. The remaining five spectra shown in Fig. 4(a) were chosen as background endmembers to generate mixed background pixels and subpixel targets. A Dirichlet distribution was employed to sample the abundances of the endmembers, ensuring compliance with nonnegative and sum-to-one constraints. Subsequently, the linear mixture model (LMM) was applied to generate samples. To simulate realistic conditions, the signal-to-noise ratio of the spectra was set to 30 dB. Each dataset comprised 1000 bags, with 300 positive bags, and 700 negative bags. In each positive bag, there were five target subpixels. We created three simulated datasets, each with mean target proportions $\bar{p}_t$ of 0.1, 0.2, and 0.3, respectively.

b) MUUFL Gulfport hyperspectral dataset: This dataset [47][1] was collected via CASI-1500 airborne hyperspectral sensor over the University of Southern Mississippi campus in November 2010. The dataset consists of five hyperspectral images, each containing approximately 330 × 340 pixels with a spatial resolution of 1 m. The images encompass 72 bands,

[1]https://github.com/GatorSense/MUUFLGulfport

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TABLE I
OVERVIEW OF TARGET TYPES AND COUNTS IN REAL-WORLD HYPERSPECTRAL DATASETS

Dataset	Target Type	Num. of Targets by Size (m)			Num.
		0.5×0.5	1×1	3×3	
MUULF Gulfport [47]	Brown	5	5	5	15
	Dark Green	5	5	5	15
	Faux Vineyard Green	4	4	4	12
	Pea Green	5	5	5	15
Avon [48]	Brown	-			20
	Blue	-			12

ranging from 375- to 1050-nm wavelengths, with a spectral sampling interval of approximately 10 nm. We excluded eight noisy bands (bands 1–4 and bands 69–72) from the analysis. In the campus area, four cloth panels of different colors and sizes were positioned as the targets of interest, as described in Table I. Each type of target had three different sizes (0.25, 1, and 9 m^2) and could potentially be occluded by oak trees, posing a challenging subpixel target detection task. Fig. 4(b) provides a visual representation of the ground-truth target locations.

We designed two detection tasks on this dataset to evaluate our method: single-type target detection (STTD) and multiple-type target detection (MTTD). In the STTD task, three hyperspectral images, Flights 1–3, were selected for training. Two images, Flights 4 and 5, were used for validation and testing, respectively. Data bags were constructed based on GPS information, which has an accuracy of approximately 1–2 m. For each target, a 5×5 square region centered around the GPS coordinates was defined as a positive bag to ensure the target was included. Areas without any targets were partitioned into negative bags. As for MTTD, we combined four types of targets and set up a multiple-type target detection configuration. In each training hyperspectral image, we created 57 positive bags. Similarly, the remaining background area was divided into negative bags.

c) Avon HTD dataset: The Avon dataset [48][2] was acquired using the ProSpecTIR airborne hyperspectral sensor over Avon, a small town located in the southern region of Rochester, New York, USA. It provides a spatial resolution of approximately 1 m and consists of 360 spectral bands ranging from 400- to 2450-nm wavelengths. The pseudo-color image of the Avon dataset is shown in Fig. 4(c), which spans 1635×302 pixels. The scene encompasses two target types: 20 brown fabric panels and 12 blue fabric panels, as illustrated in Fig. 4(d) and Table I. The precise locations of these targets were determined using the wildfire airborne sensor program (WASP), which employs a KCM-11 visible camera. WASP not only enables accurate geo-referencing but also facilitates the generation of high-resolution ground imagery within the defined target area.

In the detection experiments, two images (Avon-AM and Avon-PM), captured in the morning and afternoon of the same day, were selected for cross-validation. A 5×5 square region was defined around each target to create a positive bag, while the remaining target-free areas were divided into negative bags.

d) AeroRIT hyperspectral dataset: The AeroRIT dataset [49] was captured over the Rochester Institute of Technology's university campus using a Headwall Photonics Micro Hyperspectral E-Series CMOS sensor. The dataset has a spatial size of 1973×3975 pixels with a spatial resolution of 0.4 m. The spectral range spans from 397 to 1003 nm, consisting of 372 bands in total. For this study, every tenth band between 400 and 900 nm is sampled, resulting in 51 spectral bands, consistent with [49]. As shown in Fig. 4(e), the scene contains numerous cars, which serve as the targets of interest. The locations of these cars are indicated in Fig. 4(f). Similar to [49], the dataset is divided into three parts: left (1973×1735), middle (1973×512), and right (1973×1728), which are used for training, validation, and testing, respectively. We divide the training image into 25×25 regions, where regions containing car targets are labeled as positive bags, resulting in 5056 positive examples. The remaining regions without targets are labeled as negative bags.

2) Implementation Details: The proposed PE-PMN algorithm was implemented using Python 3.9.18 and Torch 1.13.1. All experiments pertaining to the PE-PMN algorithm were conducted using an NVIDIA GeForce RTX 3090 GPU. During the training phase, the learning rate $lr = 1 \times 10^{-3}$ was employed to optimize the model's performance, facilitating efficient convergence. To handle the training data effectively, a batch size b of 256 was utilized. $\beta = 1$ for Beta distribution in BMU. The algorithm incorporated 64 target prototypes and 64 background prototypes to robustly represent diverse target classes and background regions, respectively. To balance the contributions of different loss terms, the tuning factor $\lambda_1 = 0.5$ was applied to the PB loss $\mathcal{L}_{\text{PB}}$, while the tuning factor $\lambda_2 = 0.2$ was used for the PE loss $\mathcal{L}_{\text{Proxy}}$. In $\mathcal{L}_{\text{Proxy}}$, we set the scaling factor $\kappa = 32$, and the margin $\epsilon = 0.1$.

3) Comparison Methods: We conducted a comprehensive comparison of our PE-PMN with 15 traditional target detectors and weakly supervised techniques in total. Specifically, we included seven TC-based methods (EM-DD [11], MILES [50], MI-ACE [2], MI-SMF [2], MI-HE [6], MIMD-ACE [4], and MIMD-SMF [4]), four MINN approaches (Attention-DMIL [7], Max-attention MINN [8], CS-attention MINN [51], and MIML-Net [14]), and four traditional hyperspectral target detectors (ACE [16], SMF [15], HSD [17], and GODSNCR [24] detector) as comparison algorithms.

4) Evaluation Metrics: To evaluate the performance of our approach, we employed receiver operating characteristic (ROC) curves. The ROC curves are constructed with the false-positive rate (FPR) on the x-axis and the true-positive rate (TPR) on the y-axis. These curves allow us to assess the detection accuracy of our method in discriminating between positive and negative instances. To provide a quantitative analysis, we computed the area under the curve (AUC) for simulated datasets and the AeroRIT dataset, which provides a numerical measure of the overall performance. For real-filed MUUFL Gulfport and Avon datasets,

[2]https://www.rit.edu/cos/share2012/downloadarea.php

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TABLE II

AREA UNDER THE ROC CURVES (AUCS) FOR THE PE-PMN AND COMPARISON ALGORITHMS ON SIMULATED DATASETS WITH DIFFERENT MEAN TARGET PROPORTIONS $\bar{p}_t$ FROM 0.1 TO 0.3. BEST RESULT IN BOLD

Alg.	Weakly	DL-based	$\bar{p}_t$			Avg.
			0.1	0.2	0.3	
ACE [16]+Prior Sig.	✗	✗	0.668	0.746	0.835	0.750
SMF [15]+Prior Sig.	✗	✗	0.621	0.710	0.803	0.711
HSD [17]+Prior Sig.	✗	✗	0.664	0.738	0.810	0.737
GODSNCR [24]+Prior Sig.	✗	✗	0.701	0.753	0.814	0.756
EM-DD [11]	✓	✗	0.658	0.742	0.834	0.745
MI-ACE [2]	✓	✗	0.671	0.752	0.839	0.754
MI-SMF [2]	✓	✗	0.672	0.753	0.840	0.755
MILES [50]	✓	✗	0.744	0.775	0.786	0.768
MI-HE [6]	✓	✗	0.707	0.778	0.832	0.772
MIMD-ACE [4]	✓	✗	0.668	0.751	0.837	0.752
MIMD-SMF [4]	✓	✗	0.660	0.737	0.829	0.742
Max-attention MINN [8]	✓	✓	0.751	0.799	0.812	0.787
CS-attention MINN [51]	✓	✓	0.758	0.835	0.847	0.813
Attention-DMIL [7]	✓	✓	0.737	0.824	0.836	0.799
MIML-Net [14]	✓	✓	0.786	0.857	0.882	0.841
PE-PMN (proposed)	✓	✓	**0.804**	**0.879**	**0.902**	**0.862**

TABLE III

NAUCS AT FPR $= 1 \times 10^{-3}$ OF SINGLE-TYPE TARGET DETECTION FOR PE-PMN AND COMPARISON ALGORITHMS ON THE GULFPORT DATASET. BEST RESULT IN BOLD

Alg.	Weakly	DL-based	Target Type				Avg.
			Brown	Dark Green	Faux Vineyard Green	Pea Green	
ACE [16]+Prior Sig.	✗	✗	0.572	0.598	0.243	0.266	0.420
SMF [15]+Prior Sig.	✗	✗	0.481	0.495	0.222	0.266	0.366
HSD [17]+Prior Sig.	✗	✗	0.441	0.483	0.248	0.182	0.339
GODSNCR [24]+Prior Sig.	✗	✗	0.438	0.440	0.234	0.181	0.323
EM-DD [11]	✓	✗	0.453	0.449	0.207	0.042	0.288
MI-ACE [2]	✓	✗	0.521	0.572	0.397	0.285	0.444
MI-SMF [2]	✓	✗	0.520	0.549	0.481	0.267	0.454
MILES [50]	✓	✗	0.199	0.226	0.275	0.077	0.194
MI-HE [6]	✓	✗	0.510	0.548	0.333	0.302	0.423
MIMD-ACE [4]	✓	✗	0.520	0.495	0.159	0.298	0.368
MIMD-SMF [4]	✓	✗	0.497	0.592	0.420	0.247	0.439
Max-attention MINN [8]	✓	✓	0.580	0.542	0.227	0.302	0.413
CS-attention MINN [51]	✓	✓	0.498	0.546	0.326	0.317	0.422
Attention-DMIL [7]	✓	✓	0.440	0.576	0.347	0.283	0.412
MIML-Net [14]	✓	✓	0.587	0.675	0.503	0.370	0.534
PE-PMN (proposed)	✓	✓	**0.611**	**0.688**	**0.537**	**0.401**	**0.559**

we provided the normalized AUC values (NAUC)[3] at 1 $\times$ 10^{-3} FPR as in the previous MIL HTD techniques [2], [4], [5], [6], which focuses on evaluating the algorithm's capability to accurately identify targets while keeping false alarms at a low level.

B. Experimental Results

The AUC/NAUC metrics for the proposed PE-PMN and the comparison methods on three simulated datasets and three real-world datasets are listed in Tables II–VII, respectively. The best result is in bold. The AUC/NAUC values for the PE-PMN are consistently higher across both simulated and real-world datasets when compared to the classic and SOTA methods. Specifically, hyperspectral target detectors (ACE, SMF, HSD, and GODSNCR) that rely on predefined target spectral signatures exhibit relatively poor performance, especially on the AeroRIT dataset. This limitation arises from the substantial diversity of targets (e.g., cars) within the scene, where a single spectral prior fails to account for the intraclass variations among target objects, thereby leading to suboptimal detection accuracy. TC-based methods demonstrate slight improvements over traditional target detectors by adaptively learning discriminative spectral signatures. However, compared to deep models (MINN family), their feature representation capacity remains limited, which hinders their ability to fully capture the spectral complexities of the scene.

Previous attention-based MINN methods demonstrated notable superiority in simulated datasets, but their advantage was limited to real-world datasets. This discrepancy arises

[3]https://github.com/GatorSense/MUUFLGulfport/tree/master/MUUFLGulfportDataCollection/Bullwinkle

TABLE IV

NAUCS AT FPR $= 1 \times 10^{-3}$ OF MULTIPLE-TYPE TARGET DETECTION FOR PE-PMN AND COMPARISON ALGORITHMS ON THE GULFPORT DATASET. BEST RESULT IN BOLD

Alg.	NAUC	Alg.	NAUC
EM-DD [11]	0.256	MIMD-SMF [4]	0.272
MI-ACE [2]	0.282	Max-attention MINN [8]	0.166
MI-SMF [2]	0.226	CS-attention MINN [51]	0.266
MILES [50]	0.162	Attention-DMIL [7]	0.315
MI-HE [6]	0.279	MIML-Net [14]	0.423
MIMD-ACE [4]	0.246	**PE-PMN** (proposed)	**0.434**

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TABLE V

NAUCs AT FPR $= 1 \times 10^{-3}$ OF THE PE-PMN AND COMPARISON ALGORITHMS ON THE AVON DATASET. TRAINING ON AVON-AM, AND TEST ON AVON-PM. BEST RESULT IN BOLD

Alg.	Weakly	DL-based	Target Type		Avg.
			Brown	Blue	
ACE [16]+Prior Sig.	✗	✗	0.996	0.996	0.996
SMF [15]+Prior Sig.	✗	✗	0.996	0.997	0.997
HSD [17]+Prior Sig.	✗	✗	0.901	0.995	0.948
GODSNCR [24]+Prior Sig.	✗	✗	0.996	0.997	0.997
EM-DD [11]	✓	✗	0.805	0.988	0.897
MI-ACE [2]	✓	✗	0.994	0.995	0.995
MI-SMF [2]	✓	✗	0.994	0.996	0.995
MILES [50]	✓	✗	0.828	0.611	0.720
MI-HE [6]	✓	✗	0.907	0.994	0.951
MIMD-ACE [4]	✓	✗	0.986	0.938	0.962
MIMD-SMF [4]	✓	✗	0.991	0.995	0.993
Max-attention MINN [8]	✓	✓	0.990	0.994	0.992
CS-attention MINN [51]	✓	✓	0.996	0.996	0.996
Attention-DMIL [7]	✓	✓	0.901	0.885	0.878
MIML-Net [14]	✓	✓	0.994	0.990	0.992
PE-PMN (proposed)	✓	✓	**0.999**	**0.999**	**0.999**

TABLE VI

NAUCs AT FPR $= 1 \times 10^{-3}$ OF THE PE-PMN AND COMPARISON ALGORITHMS ON THE AVON DATASET. TRAINING ON AVON-PM, AND TEST ON AVON-AM. BEST RESULT IN BOLD

Alg.	Weakly	DL-based	Target Type		Avg.
			Brown	Blue	
ACE [16]+Prior Sig.	✗	✗	0.998	0.995	0.997
SMF [15]+Prior Sig.	✗	✗	0.998	**0.998**	0.998
HSD [17]+Prior Sig.	✗	✗	0.995	0.993	0.994
GODSNCR [24]+Prior Sig.	✗	✗	0.998	**0.998**	0.998
EM-DD [11]	✓	✗	0.789	0.855	0.822
MI-ACE [2]	✓	✗	0.998	0.993	0.996
MI-SMF [2]	✓	✗	0.996	**0.998**	0.997
MILES [50]	✓	✗	0.323	0.848	0.586
MI-HE [6]	✓	✗	0.994	0.997	0.996
MIMD-ACE [4]	✓	✗	0.997	0.973	0.985
MIMD-SMF [4]	✓	✗	0.998	**0.998**	0.998
Max-attention MINN [8]	✓	✓	0.996	0.991	0.994
CS-attention MINN [51]	✓	✓	0.997	**0.998**	0.998
Attention-DMIL [7]	✓	✓	0.995	0.993	0.994
MIML-Net [14]	✓	✓	0.996	0.993	0.995
PE-PMN (proposed)	✓	✓	**0.999**	**0.998**	**0.999**

TABLE VII

AREA UNDER THE ROC CURVES (AUCS) FOR THE PE-PMN AND COMPARISON ALGORITHMS ON THE AERORIT DATASET. BEST RESULT IN BOLD

Alg.	Weakly	DL-based	AUC
ACE [16]+Prior Sig.	✗	✗	0.792
SMF [15]+Prior Sig.	✗	✗	0.791
HSD [17]+Prior Sig.	✗	✗	0.784
GODSNCR [24]+Prior Sig.	✗	✗	0.801
EM-DD [11]	✓	✗	0.790
MI-ACE [2]	✓	✗	0.829
MI-SMF [2]	✓	✗	0.810
MILES [50]	✓	✗	0.799
MI-HE [6]	✓	✗	0.812
MIMD-ACE [4]	✓	✗	0.834
MIMD-SMF [4]	✓	✗	0.821
Max-attention MINN [8]	✓	✓	0.880
CS-attention MINN [51]	✓	✓	0.878
Attention-DMIL [7]	✓	✓	0.872
MIML-Net [14]	✓	✓	0.889
PE-PMN (proposed)	✓	✓	**0.905**

because simulated data lack severe class imbalance, whereas real-world data typically feature a much lower proportion of target data compared to background data. This imbalance hinders deep networks' ability to learn diverse target types

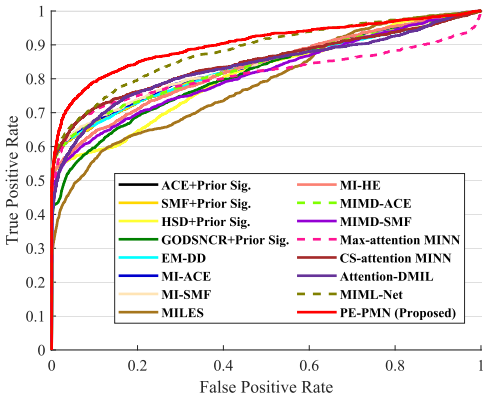

Fig. 5. ROC curves for the PE-PMN and comparison methods on a simulated dataset with mean target proportion $\bar{p}_t = 0.3$.

effectively. Our PE-PMN method addresses this challenge by employing the BMU strategy to generate samples, which helps mitigate the imbalance issue, leading to significant performance improvements. By creating additional representative samples, BMU allows the model to better understand and differentiate between targets and backgrounds, thus enhancing detection accuracy. More analysis of BMU could be found in Section IV-C.

Compared to Attention-DMIL, MIML-Net incorporates additional metric learning components that consider spectral variability and the separation between target and background. By using metric learning, MIML-Net constrains the spectral characteristics of targets and backgrounds into two distinct, compact clusters. This approach enhances performance by ensuring that the model can effectively differentiate between the two classes. The PE-PMN builds on these strengths by adopting a proxy-based method to reinforce the features of both targets and backgrounds. Additionally, it establishes learnable prototype memory banks for targets and backgrounds, which helps address the complex land-cover mixtures encountered in real-world scenarios. TC-based methods, such as MI-HE, MIMD-ACE, and MIMD-SMF, are also capable of learning MT concepts. However, they focus on learning spectral signatures of fixed dimensions (usually the number of bands). In contrast, our approach emphasizes learning sets of target and background prototype features with variable quantities and dimensions, allowing the model to better capture and adapt to variations within the data.

We present the ROC curves of the PE-PMN on both simulated datasets and the real-world MUUFL Gulfport dataset, as shown in Figs. 5–7. The ROC curves for our algorithm are mainly located near the upper left corner of the graphs. This demonstrates the PE-PMN's ability to accurately identify targets while keeping false alarm rates relatively low. The superior placement of our ROC curves highlights the algorithm's effective balance of target detection and background suppression, ensuring robust performance across diverse detection scenarios.

C. Ablation Study

To validate the effectiveness of our proposed BMU, we conducted a series of ablation studies. We evaluated the

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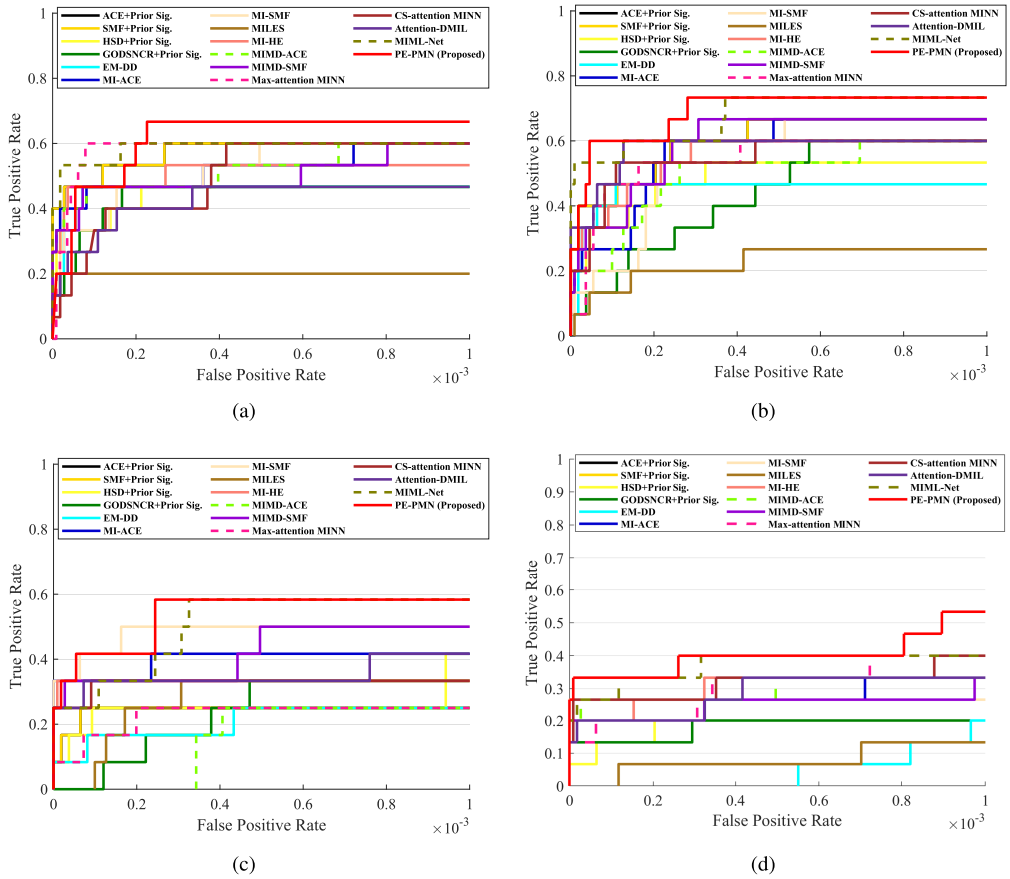

Fig. 6. ROC curves at 1×10^{-3} FPR for (a) brown, (b) dark green, (c) faux vineyard green, and (d) pea green targets.

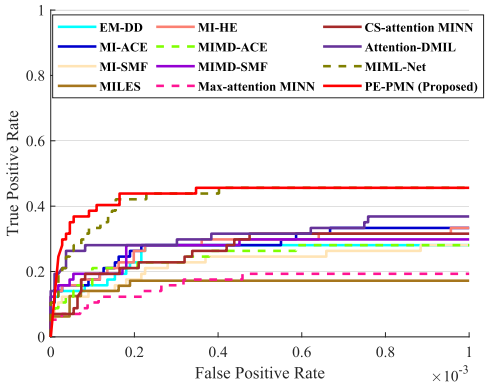

Fig. 7. ROC curves at 1×10^{-3} FPR for multiple-type target detection.

performance of several representative algorithms, including EM-DD, MI-ACE, MIMD-ACE, Attention-DMIL, and the proposed PE-PMN, both with and without the BMU, on two real-world datasets. The experimental results are shown in Table VIII.

The TC-based method EM-DD showed limited gains across the three datasets, because EM-DD tends to find the most likely target spectrum within the existing data during the E-step, and the mixed data generated by BMU has little impact on this process. In contrast, MI-ACE learns features that yield high values in spectral bands where the target spectrum and background materials differ significantly, and low values where their responses are similar. The BMU can generate complex mixed spectra, aiding MI-ACE in learning more discriminative features. MIMD-ACE, with its ability to learn various subtypes of targets, also benefits from the additional meaningful samples provided by BMU. DL-based methods such as Attention-DMIL and our proposed PE-PMN exhibit significant improvements with the inclusion of the BMU. This enhancement is particularly pronounced for Attention-DMIL. Additionally, we observed that the enhancements were particularly notable in the MUULF Gulfport dataset, where the BMU strategy contributed to better handling of complex data patterns and effectively addressed target recognition problems with partial occlusion.

The ablation experiments for PB loss and PE loss, which are discussed in detail in Section IV-D, further demonstrate the effectiveness of these components.

D. Parameter Sensitivity

The parameter sensitivity experiments are conducted on one simulated dataset with a mean target proportion of 0.3, and one real-field MUUFL Gulfport dataset.

1) Number of Prototypes M_t and M_b: To analyze the impact of the number of target prototypes M_t and the number of background prototypes M_b on performance, we conducted 2-D parameter sensitivity experiments, as presented in Fig. 8(a) and (b). First, we varied the number of target

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TABLE VIII

IMPACT OF THE BMU MODULE ON ALGORITHM PERFORMANCE (✗: WITHOUT BMU, ✓: WITH BMU), ↑:IMPROVED, ↓: DECLINED

Dataset	Target Type	EM-DD[11]		MI-ACE[2]		MIMD-ACE[4]		Attention-DMIL[7]		**PE-PMN** (proposed)	
		✗	✓	✗	✓	✗	✓	✗	✓	✗	✓
MUULF Gulfport [47]	Brown	0.453	0.459 ↑	0.521	0.530 ↑	0.520	0.536 ↑	0.440	0.512 ↑	0.579	0.611 ↑
	Dark Green	0.449	0.452 ↑	0.572	0.579 ↑	0.495	0.529 ↑	0.576	0.571 ↓	0.661	0.688 ↑
	Faux Vineyard Green	0.207	0.193 ↓	0.397	0.369 ↓	0.159	0.336 ↑	0.347	0.416 ↑	0.512	0.537 ↑
	Pea Green	0.042	0.103 ↑	0.285	0.301 ↑	0.298	0.291 ↓	0.283	0.329 ↑	0.376	0.401 ↑
Avon-PM [48]	Brown	0.805	0.812 ↑	0.994	0.996 ↑	0.986	0.993 ↑	0.901	0.956 ↑	0.991	0.999 ↑
	Blue	0.998	0.998	0.995	0.995	0.938	0.952 ↑	0.885	0.928 ↑	0.999	0.999
Avon-AM [48]	Brown	0.789	0.792 ↑	0.998	0.997 ↓	0.997	0.997	0.995	0.996 ↑	0.998	0.999 ↑
	Blue	0.855	0.871 ↑	0.993	0.993	0.973	0.981 ↑	0.993	0.996 ↑	0.995	0.998 ↑

prototypes while keeping the number of background prototypes constant. We observed that increasing the number of target prototypes generally improved detection performance up to a certain point, beyond which the performance plateaued or slightly declined. This trend suggests that a sufficient number of prototypes is crucial for capturing the diversity within target subtypes, but excessive prototypes might lead to redundancy or overfitting. Similarly, we varied the number of background prototypes while keeping the number of target prototypes constant. The performance also improved with an increasing number of background prototypes, indicating that capturing the variability within background patterns is essential for accurate target detection. However, as with target prototypes, too many background prototypes can result in diminishing returns. From the interaction analysis between the two parameters, we found that an optimal balance between the number of target prototypes and background prototypes is crucial. A higher number of target prototypes requires a correspondingly higher number of background prototypes to maintain performance. This balance ensures that both target and background variability are adequately modeled, enhancing the discrimination between targets and backgrounds. The PE-PMN's performance is relatively stable when the values of M_t and M_b are within the range of [32, 128].

2) Balancing Parameters λ_1 and λ_2: To evaluate the impact of the coefficients λ_1 and λ_2 for PB loss and PE loss, respectively, we conducted experiments with varying values for these coefficients, as shown in Fig. 8(c)–(f). We observed that an appropriate coefficient λ_1 for PB loss significantly enhances the model's ability to capture target and background subtypes. When the PB loss coefficient was set too low, the model struggled to adequately represent the variability within these subtypes, leading to diminished detection performance. Conversely, increasing the PB loss coefficient resulted in substantial improvements. However, if the PB loss coefficient is set too high, the model may become too specific to the training data, thereby reducing its generalization capability. This overfitting phenomenon ultimately led to a decline in performance. A value range of [0.2, 0.8] is appropriate for λ_1.

The PE loss aims to improve the discriminability of target and background features. We adjusted the PE loss coefficient λ_2 and found that λ_2 demonstrated stable performance within the range of 0.2–0.5. When the coefficient was set too low, the enhancement was insufficient to create distinct and compact feature representations, leading to lower detection accuracy. Conversely, setting the coefficient too high excessively constrained the feature representation space, which resulted in a low AUC performance.

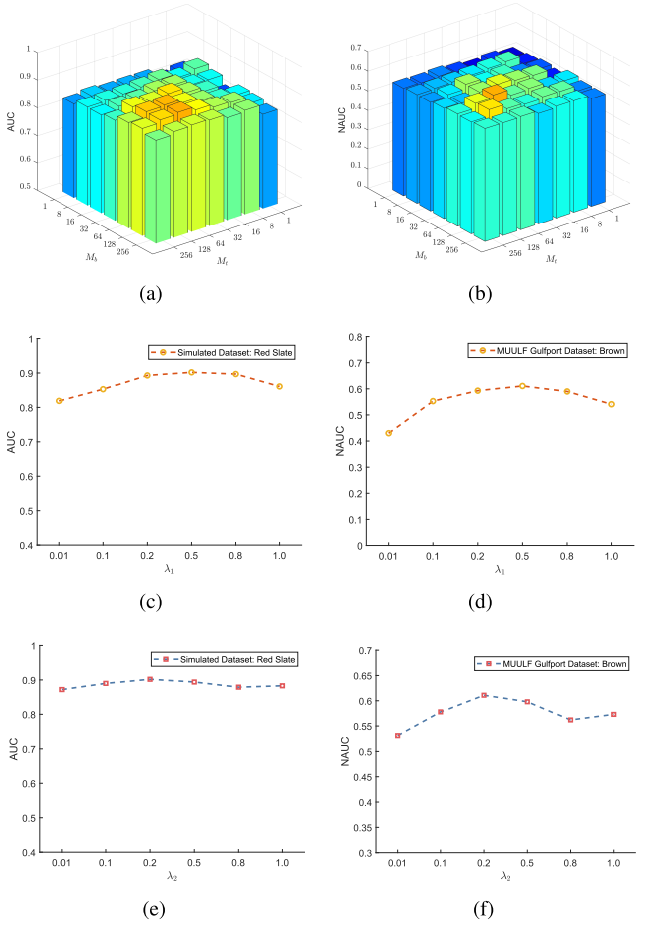

Fig. 8. Parameter sensitivity analysis. (a) Grid-search experimental results for the number of prototypes M_t and M_b on the simulated dataset with mean target proportion $\bar{p}_t = 0.3$. (b) Grid-search experimental results for the number of prototypes M_t and M_b on MUULF Gulfport dataset, brown targets. (c) Performance of the PE-PMN as λ_1 increase on the simulated dataset with mean target proportion $\bar{p}_t = 0.3$. (d) Performance of the PE-PMN as λ_1 increase on the MUULF Gulfport dataset, brown targets. (e) Performance of the PE-PMN as λ_2 increase on the simulated dataset with mean target proportion $\bar{p}_t = 0.3$. (f) Performance of the PE-PMN as λ_2 increase on the MUULF Gulfport dataset, brown targets.

V. CONCLUSION

This work tackles key challenges in HTD, including acquiring high-quality target spectra, capturing target spectral diversity, and addressing sample imbalance. We introduced

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a PE-PMN as a novel solution for weakly supervised HTD. The PE-PMN method reduces the dependency on high-quality target spectra or labor-intensive pixel-level annotations and instead leverages easily accessible region-level labels. By employing two memory prototype networks, we effectively extract and store fundamental prototypes for both targets and backgrounds. Our PE approach further enriches these prototypes and improves the differentiation between target and background features. Additionally, the BMU technique helps mitigate sample imbalance, enhancing model robustness. Our future work will explore integrating additional context information from multisource data to improve detection accuracy.

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